

Josias DJOSSOU-DAN

Software engineer

I am a software developer currently enrolled in a work-study program, with a passion for optimization and algorithms. I am currently seeking a permanent position where I can apply my technical skills to your projects.

Experiences

- 2025-09 - 2026-09

Software engineer - Apprenticeship
Siemens Mobility, transportation and rail industry
Improving the performance of an equipment simulation application in a test environment:
 - Implementation of an automated linting pipeline using LLVM Clang Tidy & Format to validate coding standards and detect quality issues.
 - Identifying bottlenecks using profiling tools (Valgrind/Callgrind)
 - implementing custom tracking classes (performance logs)
 - Implementation of a multi-threading architecture to parallelize initialization processes and maximize CPU resource utilization
 - Analysis and rewriting of critical functions to reduce time complexity and drastically speed up processing times.
 - Conducting non-regression testing to ensure the integrity of functionality following structural changes.
 - An 85% reduction in application startup time, significantly improving the user experience and operational efficiency.
- 2025-05 - 2025-09

Software engineer - Internship
Siemens Mobility, transportation and rail industry
 - Development of a log analysis tool using Qt
 - Development of modular components for real-time data processing and analysis.
 - Optimizing the interface to improve responsiveness.
 - Extension of an Event-Driven design
 - Improved usability through the implementation of a multi-window interface (dockable and detachable panels) for a flexible workspace.
 - Preparation of technical documentation and user guides
 - Set up a toolchain that includes: code compilation, lint, and formatting

Selected projects

Vehicle-to-Vehicle (V2V) Communication Simulator | C++, Qt, SQLite

- Extraction and processing of real-world geographic data (OpenStreetMap) for the interactive reconstruction of the map of Mulhouse (buildings, roads, green spaces).
- Implementation of an undirected graph to model the road network and integration of Dijkstra's algorithm for calculating optimal routes in real time.
- Design of an SQLite database for the efficient storage and retrieval of node coordinates and map metadata.
- Development of a fluid motion engine using asynchronous frame rate management (alternating calculation cycles) to minimize CPU load without compromising visual accuracy.
- Modeling of dynamic connectivity between vehicles based on the overlap of their respective radio frequencies.

Project link: https://github.com/josiashod/reseau_v2v

Simulation of Collective Behavior (Boids) | Python, Pygame

- Modeling swarm behavior (birds/fish) using autonomous agents and vector calculus.
- Implementation of separation, alignment, and cohesion rules.
- Object-oriented architecture for managing interactions.

Project link: <https://github.com/josiashod/boids>

Education

2026 | Master's degree second year, Software Science and Engineering

University of Strasbourg, Strasbourg, France

2025 | Master's Degree first year, Computer Science and Mobility

University of Haute-Alsace, Mulhouse, France

2024 | Bachelor's Degree in Computer Science

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Technical skills

Programming languages

C, C++, Python, Java, PHP, Typescript, Javascript

Fundamentals

Object-oriented design (SOLID), Data structures, Algorithm optimization, Memory management

Software development frameworks

Qt, PyQt, Swing

Software testing

Doctest, Pytest, JUnit

Methodologies

Agile, Scrum

Operating systems

Linux, Windows

Web development frameworks

React, Next.js, HTML/CSS, Laravel, Symfony, Django, Adonis.JS

Databases

MySQL, MariaDB, PostgreSQL

Tools

Git, Vs code, Qt creator, Jira, Trello, Mermaid, Modelio, Android Studio, Netbeans

Skills

Design and maintain software

Testing software

Perform continuous integration and continuous deployment (CI/CD)

Software optimization

Self-taught, Curious, Team player, Attention to detail

Languages

French

Native

English

B2

Centre d'intérêts

Climbing, calisthenics, volleyball, basketball, movies, animes